

Mahmoud Abdulbari Karsha

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Professional Summary

Software Engineer with 6+ years of experience in full-stack development, backend systems, and infrastructure.

Proven ability to build scalable web and mobile applications, manage servers, and deliver production-ready solutions.

Technical Skills

- **Languages:** TypeScript/JavaScript, Python, Dart
- **Front End Frameworks:** React, Flutter, NextJS, React Native (Expo)
- **Back End Frameworks:** NestJS, ExpressJs, FastAPI
- **Databases:** MongoDB, PostgreSQL, Redis
- **AI:** RAG systems and AI Agents
- **Infrastructure:** Ubuntu Server, Windows Server, VPS, DNS, Email Servers
- **DevOps:** Docker and Docker Compose, CI/CD (basic), Nginx and Traefik
- **Networking:** Routing, Switching and NAT.

Professional Experience

Software Engineer – NewStarters (Remote, Germany) | Feb 2023 – Present

- Built many web applications using React, NextJS, and NestJS for various clients in Germany.
- Built FastAPI service for serving AI services
- Built many Typescript libraries for internal use and open source.
- Built RAG system for document search and question answering using Langchain, OpenAI, OpenRouter and FAISS.
- Built SSO Idp and authorization server (OAuth2 and OpenID Connect).
- Built a Multi-Product microservice based platform that acts as a foundation for various applications.
- Built interactive landing pages using React and NextJS.
- Developed mobile applications using Flutter
- Developed WhatsApp chatbot

- Managed VPS infrastructure, DNS, custom cloud (Coolify), Dockers, Traefik proxy, and email servers (Mailcow).

Data Center Engineer – (Syria) | Jul 2022 – Oct 2023

- Managed enterprise servers (HPE, Dell EMC) across Linux and Windows.
- Configured Active Directory, DHCP, DNS, and Hyper-V clusters.
- Implemented storage solutions (TrueNAS) and backup systems.
- Managed networking (Cisco, Mikrotik, VPN, firewalls).

Software Developer – (Syria) | Jun 2020 – Jul 2022

- Built chat applications with file transfer and calling features.
- Developed financial and salary management systems.
- Delivered delivery mobile app with backend and dashboard.
- Optimized legacy PHP systems and improved SQL performance.
- Designed frontend layouts and backend features.
- Built hardware-integrated solutions using Raspberry Pi and Arduino.

Network Technician – Tera Center (Syria) | Feb 2017 – Apr 2020

- Managed ISP infrastructure using Mikrotik and Ubiquiti devices.
- Provided hardware and software troubleshooting.

Education

BSc in Computer Systems Engineering – Cordoba Private University (2024)

Languages

- Northern Kurdish (*Kurmanji*): native
- Central Kurdish (*Sorani*): good
- Arabic: fluent
- English: very good

Certificates

- Introduction to Software Product Management
- React Basics

Projects

- **NodeJs Http Library**: modern, TypeScript-first Node.js server framework built on Express.js. It provides a clean, modular architecture for building scalable web applications and APIs with built-in security, logging, and error handling.
- **Error Guard Library**: Advanced error transformation library for structured error management in Typescript.
- **Custom Keycloak Based Auth Library**: A comprehensive authentication and authorization package built on top of Keycloak, providing a unified API for user management, authentication, and role-based access control across multiple clients and applications.
- **Archive Project**: Electron, React and ExpressJS based application that supports searching in docs content.
- **IT Solutions Project**: React website for an IT company.